

Technical Supplement

What Does Timing Add to Size?

Probabilistic Forecasts of Recorded Cascade Growth on Two Archives

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This document accompanies the IEEE Access manuscript. It preserves the experimental scope, numerical exclusions and full supplementary results. Original experiment identifiers appear here solely to connect claims to their archived evidence. Preparation assistance is disclosed in the manuscript Acknowledgment [1]. Artifact paths in this document are relative to the companion `support/` folder.

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Scientific source version: 13 September 2026. Companion to the IEEE Access author-review manuscript. Sections S1–S7 preserve the original Phase 13F experiment and review. S8 integrates the completed Phase 14A/14B supplement; S9 records the independent review of its saved outputs.

S1 Study provenance and frozen decisions

The current experiment is a new, source-specific forecast comparison informed by earlier BRACE development. It uses a root-relative endpoint of 86,400 seconds. Older studies used an additional 24 hours after a prediction prefix. Their numerical results are not pooled with, or renamed as, the current results.

Record	Purpose	Evidence status
Legacy development through Phase 12D	Investigate count forecasting, temporal adapters and numerical behavior	Development history; different cohorts and, in relevant earlier studies, a different horizon
13A	Identify and audit the DeepHawkes Weibo release	Ingestion reproduced with documented anomalies; no model evaluation
13B	Freeze cohorts, information sets and root-relative target	26,000 roots, 78,000 windows; preparation reproduced
13C / 13C-R1	Bounded TRAIN pilot and numerical compatibility	Initial timeout retained; recovery completed; 17 original NGBoost score failures retained
13D	Score the same saved NGBoost laws under a refined certificate; bounded domain-adapter decision	1,536 score calculations completed; original failures preserved; CASPER adapter unqualified
13E	Select shared family settings using VALIDATION	154 jobs, 74 score failures; 40 qualified objects locked
13F	One evaluation of retained objects on TEST	All 600,000 model-window scores complete; six primary comparisons available
Original Phase 13F review	Check artifacts and independently aggregate saved results; write manuscript	No model loads, fits, forecasts, row scores or bootstrap draws
Phase 14A supplement	Stratify stored losses and compute saved-median MSLE	Mac reaggregation matches the supplied Linux reference; no new real forecasts or CRPS
Phase 14B supplement	Dimension-matched noise control	42 procedures, 48 component fits, 630,000 complete scores and 12 available supplementary contrasts
Phase 14 return review	Verify archive, saved rows, aggregates and saved bootstrap quantiles; integrate manuscript	No new fits, predictions, distribution scores or bootstrap draws

The detailed protocols are retained in `protocols/PROTOCOL_PHASE13B.md`, `protocols/PROTOCOL_PHASE13E.md`, `protocols/PROTOCOL_PHASE13F.md`, and `protocols/PROTOCOL_PHASE14.md`. The Phase 14 freeze receipt is also retained. The records are prospective with respect to their stated performance evaluations, informed by disclosed earlier results and metadata. No external registry acceptance is claimed. The hashes establish the relationship among retained artifacts; their timestamps alone are not an independent public preregistration service.

Weibo roles are assigned by root-user hash into 60/20/20 pools, followed by a separate root ordering to select 6,000/2,000/5,000 roots. SEISMIC uses the declared day blocks and prior-use exclusions. Exact identity strings are preserved, including scientific-notation IDs. Hashes and fixed cohort files, rather than floating-point conversion of identifiers, define membership. All quality exclusions were specified before current model evaluation.

S2 Exact model and selection specification

All models fit 18,000 rows from the 6,000 TRAIN roots in a source. Each root contributes the same three windows. Poisson, NB2 and Hurdle use TRAIN-fitted scaling; forest and boosting inputs retain the fixed feature transforms. Targets are stored separately from features, and the prediction contract rejects feature records containing a response field.

S2.1 Grids and fitting

Family	Ordered candidates, indices 0–3	Fixed settings
Hurdle	Count ridge 1, 0.1, 0.01, 0.001	Logistic activity gate, C=1; shifted positive count 1+NB2
NB2 GLM	Ridge 1, 0.1, 0.01, 0.001	Penalized joint likelihood, log-link mean, common size
Poisson GLM	Ridge 1, 0.1, 0.01, 0.001	Log-link mean
QRF	(depth, minimum leaf) = (8,50), (8,10), (12,50), (12,10)	128 trees; bootstrap; all predictors available per split; one worker
NGBoost	(stages, depth) = (100,1), (300,1), (100,2), (300,2)	Normal LogScore on $\log_{1p}(Y)$; leaf minimum 10; learning rate 0.03
Conditional empirical	k=250,100,50,25	Same-window nearest log-prefix size; lexical root-ID tie-break
Empirical by window	One distribution	All TRAIN responses at the corresponding window

The Hurdle gate uses lbfgs, tolerance 1e-7 and at most 1,000 iterations. NB slopes have an L2 penalty and the intercept is unpenalized. NB coefficients are bounded in $[-20,20]$, and size in $[0.001,1e6]$. L-BFGS-B limits are 1,000 iterations and 1,500 objective evaluations, with $ftol=1e-10$ and $gtol=1e-6$. Poisson uses at most 1,000 iterations and tolerance 1e-7. Constant-law branches handle degenerate counts explicitly. The Hurdle’s positive component is shifted; it is not the zero-truncated NB likelihood used in some published hurdle implementations.

For each tree, QRF weights every original TRAIN row that shares the query leaf equally, then averages the tree distributions. Bootstrap training shapes the tree, but leaf support is not restricted to bootstrap multiplicities. Its finite support is retained, including when a held-out response exceeds the largest training outcome.

NGBoost uses natural gradients, full rows and predictors at each stage, internal tolerance 1e-4, and no validation-based early stopping. Stage budget and actual returned stages are recorded. It trains a transformed continuous likelihood, not the discrete likelihood induced by rounding. The CDF in the manuscript is the exact count mapping tested by the scorer.

S2.2 Seed schedule, eligibility and selection

QRF and NGBoost use seeds 20260913, 20260914 and 20260915. Other procedures use the first seed. A shared candidate is eligible only if every required I0/I1/seed job completes its fit and all validation rows. Select the smallest average validation root CRPS across both information sets and the prescribed seeds. Differences within 1e-8 tie; the lowest candidate index wins. The conditional empirical k is selected by I0 validation CRPS.

Source	Family	Selected candidate	Setting	Retained objects
Weibo	Hurdle	0	Ridge 1	2
Weibo	NB2 GLM	0	Ridge 1	2
Weibo	Poisson GLM	3	Ridge 0.001	2
Weibo	QRF	3	Depth 12, leaf 10	6
Weibo	NGBoost	0	100 stages, depth 1	6
Weibo	Conditional empirical	3	k=25	1
Weibo	Empirical by window	0	Fixed	1
SEISMIC	Hurdle	3	Ridge 0.001	2
SEISMIC	NB2 GLM	3	Ridge 0.001	2
SEISMIC	Poisson GLM	3	Ridge 0.001	2
SEISMIC	QRF	3	Depth 12, leaf 10	6
SEISMIC	NGBoost	3	300 stages, depth 2	6
SEISMIC	Conditional empirical	1	k=100	1
SEISMIC	Empirical by window	0	Fixed	1

The 154 validation jobs comprised 144 learned procedures and ten reference preparations. Total component fits were 160. Of 924,000 validation score attempts, 923,926 completed and 74 failed the frozen numerical contract. All failures were on Weibo: 11 Hurdle, nine NB2 and 54 NGBoost scores. The 21 affected jobs had converged fits but incomplete scoring; all were retained as ineligible. The chosen 40 objects completed validation and were not refitted after selection. Their bindings are in `evidence/MODEL_LOCK.json`.

S2.3 Missing specialized process comparison

The declared CASPER adaptation did not qualify. Its final feasibility stage had 12 nonempty attempts: 11 reached the 20,000-iteration limit and one reached its time limit; six empty cases needed no optimization. No qualified predictive means or performance scores followed. Earlier development also investigated a capped TiDeH-inspired adapter under a different experiment contract. Neither branch supplies a like-for-like published process baseline for the present table. The missing baseline limits domain-method comparisons, while leaving the narrower within-family information contrast interpretable. It must not be reported as a loss by published CASPER or TiDeH.

S3 Scores, numerical rules and inference

For discrete CRPS, empirical laws use their finite-support identity. Poisson, NB2 and Hurdle use the inherited direct-body evaluation and an omitted-tail bound, with target 1e-8 and maximum grid 1,048,576. Parameter means above 1e7 trigger ineligibility rather than clipping.

The rounded transformed-Normal law uses the Phase 13D direct-body plus adaptive convex-tail calculation: direct range 4,096–1,048,576; target 1e-8; tail/body switch 0.1; at most 65,536 blocks; far index at most 2^{50} ; and 30 seconds per score. The returned numerical error bound is saved with each row. This analytic tail certificate includes a floating-point guard but is not a directed-rounding proof of every arithmetic operation. Original unsuccessful score attempts remain unsuccessful in their historical phases.

The central 90% interval score is

$$(u - l) + 20(l - y)\mathbf{1}\{y < l\} + 20(y - u)\mathbf{1}\{y > u\}.$$

Intervals use the forecast’s discrete quantiles. We report width and coverage alongside the score. Absolute error uses the predictive median, not a mean mislabeled as a median. No extra mean-based comparison is imported from the old manuscript.

PIT diagnostics use ten equal bins. For each case, the randomized PIT is constructed between $F(y-1)$ and $F(y)$, with a SHA-256 uniform based on source/root/window and salt `BRACE13E-PIT-v1`, shared across models and seeds. Nonrandomized bin mass integrates the uniform over this interval; a degenerate interval is assigned to the appropriate boundary bin by the frozen code. No-growth reliability uses ten fixed probability bins. `evidence/CALIBRATION.json` contains the aggregate and per-window bin masses, frequencies, and count/sum-forecast/sum-observed triplets for every procedure. Seed averaging does not create extra independent roots.

The test pass has 40 model jobs, each with 15,000 case forecasts, hence 600,000 model-case evaluations. After averaging windows there are 200,000 model/root summaries; averaging prescribed seed losses gives 120,000 procedure/root summaries for 24 procedures. The actual test cohort comprises 10,000 roots and 30,000 root-window cases. No full-sample procedure mean would be reported after a failed row or missing seed; in this pass all selected procedures completed.

Bootstrap details are fixed: NumPy PCG64, seed 20260916, source order Weibo then SEISMIC, lexical group order, batches of 100 and int64 sample indices. There are 10,000 vectors per source and 60,000 saved contrast values in total. Weibo has 3,467 sampled root-user groups; SEISMIC has 5,000 root units. Within each vector the root-weighted mean is the ratio of sampled root-loss sums to sampled root counts. Percentile quantiles use linear interpolation at 0.05/12 and 1-0.05/12. No additional multiplicity family is inferred from the descriptive tables.

The original index-stream SHA-256 values are retained in `evidence/BOOTSTRAP_DESIGN.json`. The original Phase 13F review checked quantiles of saved replicates without regenerating the primary bootstrap. Phase 14A draws a separate exploratory subgroup bootstrap, described in S8. The conditional uncertainty statements exclude training, tuning and source-selection uncertainty and depend on the adequacy of the declared resampling units.

S4 Full secondary results

The following tables provide the complete aggregate secondary results.

S4.1 Aggregate metrics for all procedures

Source	Procedure	Info	CRPS	No-growth Brier	Median AE	90% coverage	90% width	90% interval score
Weibo	Empirical by window	I0	126.569	0.016303	140.675	89.09%	384.333	2039.451
Weibo	Conditional empirical	I0	103.816	0.015350	122.334	83.89%	337.608	1390.895
Weibo	Poisson GLM	I0	148.384	0.016738	153.551	12.27%	31.792	2801.479
Weibo	Poisson GLM	I1	146.924	0.016733	152.123	12.38%	32.089	2770.853
Weibo	NB2 GLM	I0	123.562	0.020555	139.501	96.16%	438.037	1857.146
Weibo	NB2 GLM	I1	120.559	0.019735	139.392	94.97%	402.479	1728.436
Weibo	Hurdle	I0	124.126	0.016337	140.427	94.96%	433.792	1862.830
Weibo	Hurdle	I1	121.286	0.011839	139.699	94.99%	394.252	1756.772
Weibo	QRF	I0	100.990	0.015019	122.086	91.98%	456.030	1272.354
Weibo	QRF	I1	96.640	0.010319	117.061	92.58%	460.665	1214.982
Weibo	NGBoost	I0	111.249	0.016345	133.155	92.21%	391.677	1526.020
Weibo	NGBoost	I1	110.484	0.015661	132.550	92.51%	386.910	1483.795
SEISMIC	Empirical by window	I0	56.224	0.007972	70.209	92.00%	340.000	640.027
SEISMIC	Conditional empirical	I0	42.740	0.007895	56.737	92.16%	249.932	399.028
SEISMIC	Poisson GLM	I0	66.422	0.008068	71.567	14.59%	32.628	1162.297
SEISMIC	Poisson GLM	I1	59.117	0.008068	63.971	18.63%	31.366	1025.798
SEISMIC	NB2 GLM	I0	52.041	0.008010	67.178	94.43%	301.777	526.765
SEISMIC	NB2 GLM	I1	45.554	0.008012	60.203	93.33%	269.412	418.028
SEISMIC	Hurdle	I0	52.036	0.007973	67.233	94.43%	300.405	526.018
SEISMIC	Hurdle	I1	45.022	0.005113	59.577	94.13%	267.911	412.004
SEISMIC	QRF	I0	42.875	0.007908	57.719	92.42%	284.852	387.892
SEISMIC	QRF	I1	39.058	0.005924	51.467	93.99%	276.571	367.088
SEISMIC	NGBoost	I0	42.522	0.007992	57.259	93.30%	289.582	389.978
SEISMIC	NGBoost	I1	39.440	0.006595	52.505	92.41%	235.177	370.462

S4.2 Interval coverage at all four nominal levels

Source	Procedure	Info	50%	80%	90%	95%
SEISMIC	Empirical by window	I0	52.767%	83.427%	92.000%	96.007%
Weibo	Empirical by window	I0	50.567%	79.247%	89.093%	95.627%
SEISMIC	Conditional empirical	I0	54.607%	82.740%	92.160%	96.547%
Weibo	Conditional empirical	I0	45.693%	74.820%	83.887%	93.207%
SEISMIC	Hurdle	I0	61.953%	88.707%	94.427%	96.820%
SEISMIC	Hurdle	I1	62.287%	88.207%	94.127%	96.593%
Weibo	Hurdle	I0	71.787%	92.127%	94.960%	96.700%
Weibo	Hurdle	I1	69.633%	92.133%	94.987%	96.507%
SEISMIC	QRF	I0	54.400%	82.684%	92.420%	96.469%
SEISMIC	QRF	I1	56.736%	86.093%	93.987%	97.458%
Weibo	QRF	I0	51.324%	80.893%	91.984%	96.158%
Weibo	QRF	I1	54.231%	82.973%	92.580%	96.700%
SEISMIC	NGBoost	I0	56.700%	86.753%	93.300%	96.067%
SEISMIC	NGBoost	I1	55.640%	85.653%	92.413%	95.807%
Weibo	NGBoost	I0	53.493%	86.727%	92.213%	94.893%
Weibo	NGBoost	I1	53.467%	85.607%	92.513%	95.400%
SEISMIC	NB2 GLM	I0	62.307%	88.847%	94.433%	96.740%
SEISMIC	NB2 GLM	I1	62.240%	87.733%	93.327%	95.913%
Weibo	NB2 GLM	I0	73.800%	92.467%	96.160%	97.260%
Weibo	NB2 GLM	I1	71.347%	91.607%	94.967%	96.827%
SEISMIC	Poisson GLM	I0	6.480%	11.680%	14.593%	17.433%
SEISMIC	Poisson GLM	I1	7.887%	14.800%	18.627%	21.833%
Weibo	Poisson GLM	I0	5.440%	9.980%	12.273%	14.420%
Weibo	Poisson GLM	I1	4.820%	9.520%	12.380%	14.793%

S4.3 Window-specific primary information contrasts

Descriptive per-window summaries; no separate window-wise hypothesis tests.

Source	Family	Prefix (min)	I0 CRPS	I1 CRPS	I1 – I0	Relative reduction
Weibo	Hurdle	5	137.348	134.555	-2.793	2.03%
Weibo	Hurdle	15	128.398	125.443	-2.955	2.30%
Weibo	Hurdle	60	106.632	103.861	-2.771	2.60%
Weibo	QRF	5	119.100	113.362	-5.738	4.82%
Weibo	QRF	15	102.225	99.054	-3.172	3.10%
Weibo	QRF	60	81.646	77.505	-4.141	5.07%
Weibo	NGBoost	5	131.090	129.145	-1.945	1.48%
Weibo	NGBoost	15	113.285	113.222	-0.063	0.06%
Weibo	NGBoost	60	89.374	89.085	-0.289	0.32%
SEISMIC	Hurdle	5	62.254	55.138	-7.117	11.43%
SEISMIC	Hurdle	15	53.150	45.085	-8.064	15.17%
SEISMIC	Hurdle	60	40.704	34.843	-5.861	14.40%
SEISMIC	QRF	5	52.613	48.476	-4.137	7.86%
SEISMIC	QRF	15	43.388	38.257	-5.131	11.83%
SEISMIC	QRF	60	32.626	30.442	-2.183	6.69%
SEISMIC	NGBoost	5	52.510	49.376	-3.134	5.97%
SEISMIC	NGBoost	15	42.623	38.824	-3.800	8.91%
SEISMIC	NGBoost	60	32.434	30.120	-2.314	7.13%

The companion machine-readable tables preserve full precision:

- `tables/ALL_PROCEDURES.md`: CRPS, no-growth Brier, median absolute error, 90% coverage, width and interval score for 24 procedures.
- `tables/INTERVAL_COVERAGE.md`: nominal 50/80/90/95% coverage.
- `tables/PER_WINDOW_PRIMARY_FAMILIES.md`: all 18 descriptive timing contrasts by window.
- `tables/TEST_SAMPLE_PROFILE.md`: post-TEST descriptive response and prefix profile.
- `evidence/PROCEDURE_METRICS.json`: full-precision aggregate and per-window results, including the simple references.
- `evidence/CALIBRATION.json`: both PIT constructions and all no-growth reliability bins.

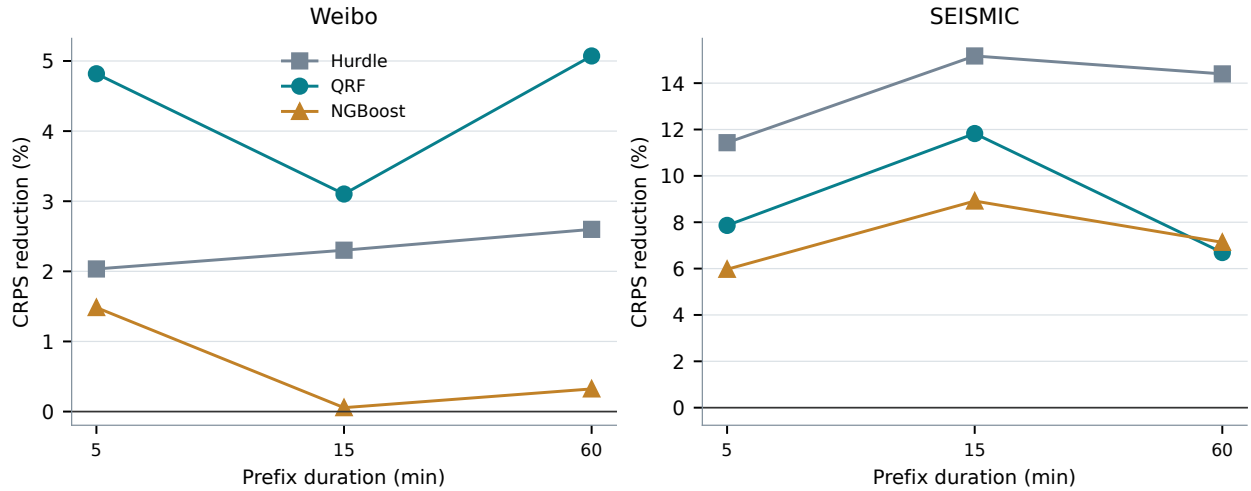


Figure S1: Per-window timing reductions. Values are redrawn from the frozen aggregate evidence.

No-growth rates over the 15,000 unique source-window cases are 251/15,000 (1.6733%) on Weibo and 121/15,000 (0.8067%) on SEISMIC. The three windows within a root are dependent. These rates help interpret the Brier scale and should not be treated as independent-event incidence estimates. Sparse high-probability reliability bins are descriptive and can be unstable.

S5 Prespecified duplicate sensitivity

Three flagged Weibo test roots are excluded only in the descriptive sensitivity. Models and primary results remain unchanged.

Family	Kept roots	I0 CRPS	I1 CRPS	Difference
Hurdle	4,997	109.570281	107.050111	-2.520170
QRF	4,997	90.725382	86.988110	-3.737272
NGBoost	4,997	97.853775	97.212399	-0.641376

The contrast directions persist, while average score magnitudes change. The sensitivity cannot reconstruct unique events, isolate the reason a flagged root has large loss, or remove training effects of repeated records. No new significance statements are attached. SEISMIC’s `unavailable_without_event_ids` flag remains an unknown status rather than a negative duplicate finding.

S6 Original Phase 13F evidence review

The supplied Phase 13F archive is 160,964,430 bytes with SHA-256:

592e185fc9a0ad96265b407af71a3e80e9d66ccb9da140a3bd32c36bfb51ce0.

All 496 manifest payload names, sizes and hashes were checked; reading every member verified its ZIP CRC. The 39 frozen source files matched the previously prepared kit; all 40 model-object hashes matched the retained lock, itself identical to the previously reviewed Phase 13E lock. The Mac report records 18 successful synthetic tests, one completed test pass, zero fits and zero selection calls.

The independent reporting calculation read all 600,000 saved score rows, verified case identities and complete window membership, and checked all nine stored metrics through both root-aggregation levels. Model/root and procedure/root values matched exactly. The largest aggregate-metric difference from a different summation order was $7.43512\text{e-}11$; the largest per-window difference was $2.95586\text{e-}12$. Primary point values, saved-replicate quantiles and duplicate-sensitivity values matched exactly. These checks are recorded in `evidence/INDEPENDENT_REVIEW.json`.

No fitted object was deserialized in this review. No model was retrained, queried or rescored, and no bootstrap draw was repeated. Consequently, verification of the saved reporting arithmetic must be distinguished from an independent reproduction of training or scoring. Calibration plots were checked visually; their saved bin calculations were not independently reconstructed row by row.

The Mac environment is recorded in `evidence/ENVIRONMENT.json`, including Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, scikit-learn 1.8.0 and NGBoost 0.5.11. Historical cross-platform refitting discrepancies remain in the legacy record. We do not claim bitwise retraining equivalence on arbitrary operating systems.

S7 Publication figure and artifact notes

The IEEE manuscript figures were redrawn from the frozen machine-readable aggregates. Vector PDFs and PNG previews are provided, with editable Python source for numerical plots and TikZ source for the methodological diagram. No model was fitted or queried, no case was rescored and no bootstrap was regenerated while preparing the IEEE package. The archived previous reporting artifacts are preserved separately.

This manuscript package is not a substitute for the full experimental archives. Reproduction of fitting and evaluation requires the source data, model source, environments and the training-selection, locked-test and supplementary-control archives. The public deposit layout and redistribution terms require author confirmation before an open-access artifact claim can be made.

S8 Post-13F supplementary analyses, Phase 14

S8.1 Status and analysis chronology

The six primary comparisons and their diagnostics had already been observed. The Phase 14 protocol, count bins and noise schedule were frozen before the new strata were calculated and before any real noise fitting. These are prospectively specified supplementary analyses of an already observed test cohort. The Mac saved-score reaggregation matched the supplied Linux reference within the frozen tolerance. The real-data

negative control completed once on Mac. Its results are integrated below without modifying the original six primary comparisons or the historical failures.

The historical Phase 13F protocol’s 2% reference is preserved as a descriptive convention, without an operational cost justification or an inferential practical-significance test. The revised main discussion uses the full effect sizes instead of dichotomizing them at that value.

S8.2 Prefix strata and uncertainty

Bins are fixed at 0, 1–2, 3–9, 10–49 and ≥ 50 nonroot events. Analyze each window within each source. Weibo resamples entire root-user clusters, then divides sampled stratum loss sums by sampled eligible-root counts; SEISMIC resamples roots. The same 2,000 vectors per source are reused across all windows, bins and families. Each root retains its windows and original seed losses. Pointwise 95% percentile intervals require at least 30 observed sampling units and at least 99% finite resamples. They do not constitute 90 additional confirmatory tests.

Source	Prefix, min	0	1–2	3–9	10–49	≥ 50
weibo	5	1,622	1,516	1,251	526	85
weibo	15	764	982	1,720	1,168	366
weibo	60	334	386	1,547	1,814	919
seismic	5	103	106	497	2,866	1,428
seismic	15	53	49	171	2,433	2,294
seismic	60	26	20	51	1,585	3,318

Relative CRPS reduction, in percent, is $100 \times (1 - I1/I0)$; negative values are deteriorations. The following table retains all families and bins. It has no fitted threshold or selection of a preferred curve.

Source	Family	Prefix, min	0	1–2	3–9	10–49	≥ 50
weibo	hurdle	5	-0.01	2.70	-0.18	0.38	5.79
weibo	hurdle	15	-0.03	3.97	10.08	-0.28	1.75
weibo	hurdle	60	0.02	8.12	10.10	10.92	-0.39
weibo	qrf	5	0.00	-0.16	-2.20	4.35	16.50
weibo	qrf	15	0.00	0.03	0.24	2.33	5.19
weibo	qrf	60	-0.01	-1.46	0.02	3.95	6.53
weibo	ngboost	5	0.37	0.21	1.40	3.12	1.07
weibo	ngboost	15	0.09	-0.00	-1.93	-0.01	0.39
weibo	ngboost	60	0.37	-1.70	-3.31	2.44	0.49
seismic	hurdle	5	25.52	-41.63	-19.16	14.12	13.38
seismic	hurdle	15	43.17	-9.56	-10.26	15.06	15.78
seismic	hurdle	60	47.17	19.36	21.57	6.13	15.51
seismic	qrf	5	0.16	0.14	-1.34	2.60	11.38
seismic	qrf	15	0.65	7.39	-0.52	1.40	15.35
seismic	qrf	60	2.03	-5.17	8.80	-1.21	8.03
seismic	ngboost	5	1.40	2.87	5.47	2.71	7.86
seismic	ngboost	15	-0.77	3.61	-0.89	0.96	11.82
seismic	ngboost	60	-0.51	5.42	5.39	-1.30	8.68

The full tables contain 360 procedure/stratum metric rows and 90 timing-contrast rows. Partition-weighted means recover all 72 procedure/window aggregates across ten metrics within 2.28×10^{-13} . All zero-prefix I1 additions equal (1,1,0,0,0,0). An empty-prefix improvement therefore concerns the fitted function, not event-time variation inside that subgroup.

S8.3 Supplementary MSLE of the saved median forecasts

We compute natural-log MSLE, $(\log_{1p}(\text{median}) - \log_{1p}(y))^2$, from saved individual-model medians, average losses over seeds and windows within root, then over roots. The table below gives Weibo values; both sources and all procedures are retained in `tables/MSLE_SAVED_MEDIANS.csv`. These values did not select or refit a procedure. A median is generally optimal for absolute error, not for squared log loss. Gneiting [2].

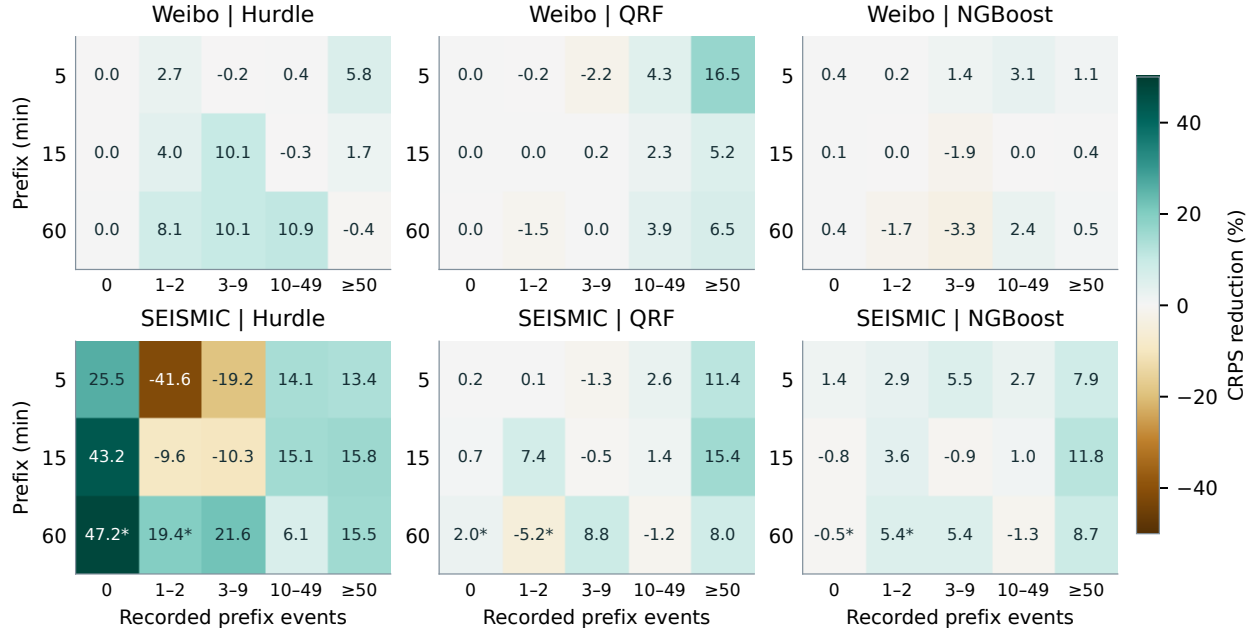


Figure S2: All source and family prefix-stratum contrasts. Values are redrawn from the frozen aggregate evidence.

Family	Information	5 min	15 min	60 min	Root-average
Empirical	I0	1.880557	2.063790	2.321336	2.088561
Conditional empirical	I0	1.323165	1.393302	1.444213	1.386894
hurdle	I0	1.539661	1.799347	2.518383	1.952464
hurdle	I1	1.534333	1.734562	2.221178	1.830024
qrf	I0	1.316971	1.404395	1.428152	1.383173
qrf	I1	1.246641	1.137481	1.101965	1.162029
ngboost	I0	1.598530	1.481395	1.544876	1.541601
ngboost	I1	1.528144	1.424909	1.500589	1.484547

These values are not directly comparable to published DeepHawkes/CasFlow leaderboards: selected roots, split rules, retained duplicates, target boundaries, information access and log conventions must first be aligned.

S8.4 Completed independent-noise control

I0 receives seven independent $\text{Uniform}(-\sqrt{3}, \sqrt{3})$ coordinates. Three fixed noise realizations are crossed with one Hurdle seed or three QRF/NGBoost seeds on each source: 42 procedures, at most 48 component fits and 630,000 model-case TEST evaluations on the existing 30,000 distinct test root-window cases. TRAIN/TEST streams are generated separately; each realization is shared across families and fitting seeds. Hyperparameters, loss definitions and numerical score rules are inherited without validation reselection. The 6,000 TRAIN roots per source match the frozen selected-model provenance byte for byte.

The root loss averages the three windows, original model seeds and three noise realizations. Twelve supplementary contrasts compare noise-I0 and I1-noise. Ten thousand common bootstrap vectors per source retain the original grouping and root weighting; bounds use 0.05/24 and 1-0.05/24, retaining the 12-slot multiplicity family when any comparison is withheld. Missing required rows, seeds or realizations withhold the whole affected noise procedure. Original I0/I1 comparisons are preserved.

Independent noise avoids treating impossible shuffled combinations as valid timing vectors, but it is not a perfect capacity match. Continuous noise and sparse, correlated timing summaries provide different opportunities for splits and shrinkage. In particular, adding noise can sometimes improve a forest's out-of-sample accuracy; the negative control is not expected to fail by definition. [3], [4], [5].

All 42 scheduled control procedures completed, using 48 component fits. Every one of the 630,000 attempted predictions and scores completed, with no prediction failures, score failures, timeouts or recorded warnings. These evaluations reuse the 10,000 test roots; they are not 630,000 independent observations. The

Mac preflight reports 17 successful unit tests and three separate synthetic procedure fits using four component fits, outside the real-data budget. No original model was refitted and no validation reselection occurred.

All 12 supplementary contrasts, full interpretation.

Source	Family	Noise CRPS	Noise - I0 [adjusted interval]	I1 - noise [adjusted interval]
Weibo	Hurdle	124.124	-0.002 [-0.036, +0.030]	-2.838 [-5.212, -1.050]
Weibo	QRF	101.944	+0.953 [-1.255, +4.803]	-5.304 [-10.532, -1.905]
Weibo	NGBoost	111.283	+0.033 [-0.036, +0.080]	-0.799 [-1.457, -0.274]
SEISMIC	Hurdle	52.063	+0.027 [-0.002, +0.056]	-7.041 [-10.573, -4.833]
SEISMIC	QRF	42.660	-0.216 [-0.515, +0.212]	-3.602 [-4.594, -2.648]
SEISMIC	NGBoost	42.697	+0.174 [-0.160, +0.793]	-3.257 [-4.145, -2.392]

Every noise-I0 interval includes zero; all I1-noise intervals are below zero. No equivalence test or margin was specified, so the first result does not prove a null augmentation effect. The second supports the predictive value of this timing representation relative to this fixed control, conditional on the retained models, realizations and sampling assumptions. The primary six-contrast family and this supplementary 12-contrast family are reported separately, without claiming joint error control over all analyses.

Separate noise realizations, mean CRPS. Each cell averages the prescribed fitting seeds and root windows; none was selected for final reporting.

Source	Family	Realization 0	Realization 1	Realization 2
Weibo	Hurdle	124.137157	124.111029	124.124508
Weibo	QRF	101.478728	101.234407	103.118272
Weibo	NGBoost	111.286162	111.249499	111.312650
SEISMIC	Hurdle	52.075040	52.067359	52.045596
SEISMIC	QRF	42.652127	42.684987	42.642352
SEISMIC	NGBoost	42.670395	42.745813	42.673987

The aggregate control averages the losses over all three realizations. It is not a predictive mixture distribution, and its bootstrap intervals do not integrate over hypothetical new noise realizations. The Weibo QRF realization means illustrate variability that remains outside those conditional intervals.

The complete records include `tables/NOISE_REALIZATION_SUMMARIES.csv`, `evidence/phase14/STAGE_B_REPORT.json`, `evidence/phase14/PROCEDURE_COMPLETENESS.json`, and both saved bootstrap arrays. The generator, settings and numerical contract were retained after all outcomes were observed.

S8.5 Scope decision and preserved history

The main paper evaluates compact summaries within general distributional learners. Specialized event-process comparisons are outside that scope. The prior numerically qualified 12D adapter and the separate unqualified CASPER adapter remain documented in the experimental history; neither establishes failure of the published methods. No new domain-process fitting or seven-feature importance sweep is scheduled in Phase 14.

The revised manuscript cites the final three-author Hooker, Mentch and Zhou (2021) journal article, *Statistics and Computing* 31, Article 82, DOI 10.1007/s11222-021-10057-z. The frozen protocol and its historical citation bytes remain unchanged.

S9 Independent review of the Phase 14 return

The received archive is `BRACE_PHASE14_RETURN_20260913T192805Z.zip`, 222,470,817 bytes, SHA-256:

5295b94b6c6bfef77b14bbe94d16ed977e489e05dafa5e0b667d49cd1f7d2784.

The review checked the inventory, safe unique paths, sizes, SHA-256 and CRC for all 521 payloads. All 108 source-manifest payloads and its manifest match the original supplementary kit byte for byte. Forty-two model files were hashed without deserialization. All 12 noise-matrix files matched their bindings.

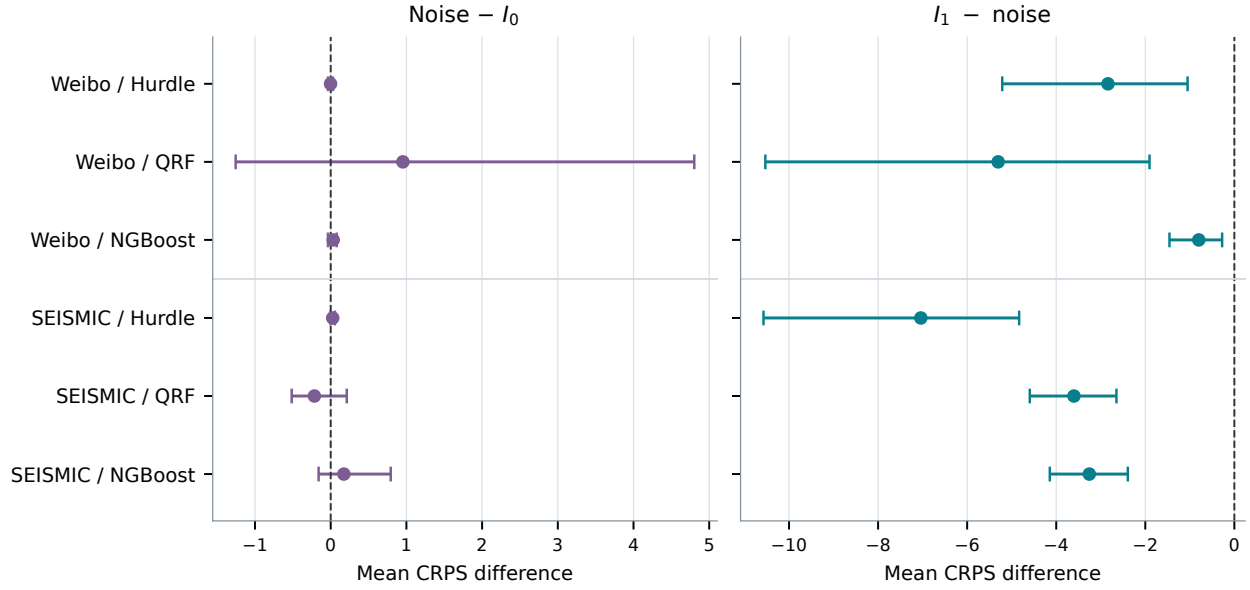


Figure S3: Supplementary noise contrasts. Values are redrawn from the frozen aggregate evidence.

An independent script read every one of the 630,000 saved score rows, checking unique root/window membership, source, group, prefix count, response, completion status and the stored CRPS error bound. It reconstructed all six noise-procedure loss arrays from the required seeds and three realizations, compared all 18 realization means and all 12 contrasts, and evaluated the declared quantiles of the saved bootstrap replicates. Regrouping the floating-point summation over realizations produced a maximum difference of 5.821×10^{-11} across the ten-metric loss arrays, within the review tolerance; the maximum difference in reported noise-contrast point quantities was 6.662×10^{-16} . All saved interval endpoints matched exactly.

The review also recovered all 90 stratum CRPS contrasts and their interval decisions from saved procedure arrays. There are 64 negative and 26 positive point estimates; these are direction counts, not counts of statistically established subgroup effects. Six intervals remain withheld for the two sparse SEISMIC strata at 60 minutes across all three families. The primary CRPS means remain consistent with the unchanged Phase 13F record, with a maximum summation-order difference of 1.422×10^{-14} .

No models were fitted or queried, no distribution scores were recomputed, and no bootstrap samples were generated during this review. The saved bootstrap quantiles were checked without regenerating the index streams; the frozen grouping implementation was inspected. This is verification of recorded evidence, not an independent rerun of training or numerical scoring. The Mac source/parent integrity receipts are retained separately from this review. The independent verifier and its bounded scope are in `code/review_saved_phase14.py` and `evidence/phase14/INDEPENDENT_PHASE14_REVIEW.json`.

S10 Complete exploratory strata and empirical references

All 90 point contrasts are retained. Negative differences favor timing. The intervals below are exploratory pointwise summaries; six are withheld under the frozen small-sample rule. Root counts and resampling-unit counts have different meanings on Weibo. They coincide on SEISMIC. Raw CRPS is source specific.

S10.1 Weibo: every exploratory stratum contrast

Family	min	Bin	Roots	Units	I_0	I_1	Δ	Pointwise 95% CI
Hurdle	5	0	1622	1438	71.885	71.892	0.007	[-0.032, 0.047]
Hurdle	5	1-2	1516	1277	54.103	52.642	-1.461	[-1.854, -1.032]
Hurdle	5	3-9	1251	934	97.159	97.336	0.177	[-1.137, 2.071]
Hurdle	5	10-49	526	362	324.786	323.562	-1.223	[-7.297, 3.118]
Hurdle	5	50+	85	67	2302.840	2169.398	-133.442	[-281.377, -26.371]
Hurdle	15	0	764	708	71.266	71.287	0.021	[-0.034, 0.080]
Hurdle	15	1-2	982	889	63.848	61.312	-2.536	[-2.979, -2.075]
Hurdle	15	3-9	1720	1357	38.055	34.217	-3.838	[-4.232, -3.437]
Hurdle	15	10-49	1168	816	98.380	98.655	0.276	[-0.946, 1.814]
Hurdle	15	50+	366	260	941.212	924.760	-16.452	[-36.451, 0.919]
Hurdle	60	0	334	316	78.529	78.509	-0.019	[-0.087, 0.055]
Hurdle	60	1-2	386	362	30.717	28.224	-2.493	[-3.198, -1.758]
Hurdle	60	3-9	1547	1312	41.215	37.054	-4.161	[-4.385, -3.921]
Hurdle	60	10-49	1814	1341	39.687	35.354	-4.333	[-4.764, -3.859]
Hurdle	60	50+	919	617	390.992	392.527	1.535	[-12.669, 12.592]
QRF	5	0	1622	1438	67.251	67.250	-0.001	[-0.002, 0.001]
QRF	5	1-2	1516	1277	45.124	45.195	0.071	[-0.050, 0.190]
QRF	5	3-9	1251	934	93.818	95.884	2.065	[-0.507, 6.371]
QRF	5	10-49	526	362	284.970	272.585	-12.385	[-17.610, -7.387]
QRF	5	50+	85	67	1773.518	1480.974	-292.544	[-571.061, -105.531]
QRF	15	0	764	708	68.067	68.065	-0.002	[-0.008, 0.004]
QRF	15	1-2	982	889	53.850	53.832	-0.018	[-0.255, 0.188]
QRF	15	3-9	1720	1357	25.636	25.575	-0.062	[-0.325, 0.125]
QRF	15	10-49	1168	816	91.928	89.785	-2.144	[-5.497, 2.775]
QRF	15	50+	366	260	696.112	659.965	-36.147	[-85.338, 47.222]
QRF	60	0	334	316	76.767	76.774	0.007	[-0.005, 0.019]
QRF	60	1-2	386	362	22.000	22.321	0.321	[0.123, 0.518]
QRF	60	3-9	1547	1312	23.796	23.791	-0.004	[-0.184, 0.211]
QRF	60	10-49	1814	1341	25.450	24.445	-1.005	[-1.817, -0.276]
QRF	60	50+	919	617	316.777	296.103	-20.675	[-43.898, 7.836]
NGBoost	5	0	1622	1438	68.044	67.790	-0.254	[-0.412, -0.088]
NGBoost	5	1-2	1516	1277	46.157	46.060	-0.097	[-0.309, 0.121]
NGBoost	5	3-9	1251	934	99.573	98.179	-1.394	[-2.180, -0.722]
NGBoost	5	10-49	526	362	341.294	330.658	-10.637	[-15.151, -6.172]
NGBoost	5	50+	85	67	2012.017	1990.545	-21.472	[-45.309, 1.277]
NGBoost	15	0	764	708	68.817	68.756	-0.061	[-0.165, 0.042]
NGBoost	15	1-2	982	889	55.542	55.544	0.002	[-0.280, 0.277]
NGBoost	15	3-9	1720	1357	26.747	27.263	0.516	[0.158, 0.862]
NGBoost	15	10-49	1168	816	96.364	96.372	0.007	[-0.796, 0.828]
NGBoost	15	50+	366	260	821.718	818.531	-3.186	[-10.285, 3.449]
NGBoost	60	0	334	316	79.626	79.333	-0.293	[-0.413, -0.182]
NGBoost	60	1-2	386	362	22.165	22.542	0.377	[0.085, 0.651]
NGBoost	60	3-9	1547	1312	25.492	26.335	0.843	[0.626, 1.060]
NGBoost	60	10-49	1814	1341	27.293	26.628	-0.665	[-1.026, -0.326]
NGBoost	60	50+	919	617	351.220	349.491	-1.729	[-6.235, 2.497]

S10.2 SEISMIC: every exploratory stratum contrast

Family	min	Bin	Roots	Units	I_0	I_1	Δ	Pointwise 95% CI
Hurdle	5	0	103	103	53.193	39.615	-13.577	[-18.972, -8.030]
Hurdle	5	1-2	106	106	58.056	82.226	24.169	[14.453, 34.600]
Hurdle	5	3-9	497	497	25.328	30.181	4.853	[2.747, 7.191]
Hurdle	5	10-49	2866	2866	35.251	30.274	-4.977	[-5.481, -4.449]
Hurdle	5	50+	1428	1428	130.266	112.834	-17.433	[-26.817, -9.793]
Hurdle	15	0	53	53	64.929	36.898	-28.031	[-36.153, -19.826]
Hurdle	15	1-2	49	49	53.260	58.349	5.089	[-20.517, 30.204]
Hurdle	15	3-9	171	171	31.900	35.172	3.273	[0.603, 6.455]
Hurdle	15	10-49	2433	2433	26.725	22.699	-4.026	[-4.426, -3.610]
Hurdle	15	50+	2294	2294	82.485	69.473	-13.012	[-18.032, -9.267]

Family	min	Bin	Roots	Units	I_0	I_1	Δ	Pointwise 95% CI
Hurdle	60	0	26	26	84.581	44.688	-39.893	withheld
Hurdle	60	1-2	20	20	48.631	39.216	-9.415	withheld
Hurdle	60	3-9	51	51	37.352	29.294	-8.058	[-15.475, -1.542]
Hurdle	60	10-49	1585	1585	20.957	19.672	-1.284	[-1.855, -0.614]
Hurdle	60	50+	3318	3318	49.797	42.072	-7.725	[-10.022, -5.975]
QRF	5	0	103	103	36.145	36.086	-0.059	[-0.113, -0.004]
QRF	5	1-2	106	106	50.433	50.362	-0.071	[-0.955, 0.811]
QRF	5	3-9	497	497	20.612	20.888	0.276	[0.057, 0.482]
QRF	5	10-49	2866	2866	27.535	26.818	-0.717	[-0.937, -0.505]
QRF	5	50+	1428	1428	115.430	102.298	-13.132	[-16.518, -9.852]
QRF	15	0	53	53	34.901	34.673	-0.228	[-0.642, 0.122]
QRF	15	1-2	49	49	40.560	37.565	-2.996	[-5.189, -0.498]
QRF	15	3-9	171	171	25.944	26.080	0.135	[-0.646, 0.929]
QRF	15	10-49	2433	2433	19.192	18.924	-0.268	[-0.476, -0.063]
QRF	15	50+	2294	2294	70.606	59.766	-10.841	[-13.238, -8.473]
QRF	60	0	26	26	43.816	42.926	-0.889	withheld
QRF	60	1-2	20	20	19.619	20.634	1.015	withheld
QRF	60	3-9	51	51	24.130	22.006	-2.124	[-4.222, 0.117]
QRF	60	10-49	1585	1585	14.122	14.292	0.170	[-0.083, 0.426]
QRF	60	50+	3318	3318	41.587	38.249	-3.338	[-4.800, -1.756]
NGBoost	5	0	103	103	37.285	36.764	-0.521	[-1.341, 0.300]
NGBoost	5	1-2	106	106	52.904	51.387	-1.517	[-2.370, -0.613]
NGBoost	5	3-9	497	497	23.478	22.194	-1.284	[-1.662, -0.911]
NGBoost	5	10-49	2866	2866	28.121	27.359	-0.762	[-0.960, -0.558]
NGBoost	5	50+	1428	1428	112.633	103.784	-8.849	[-11.661, -5.925]
NGBoost	15	0	53	53	34.506	34.771	0.265	[-0.475, 0.969]
NGBoost	15	1-2	49	49	40.048	38.602	-1.446	[-2.519, -0.466]
NGBoost	15	3-9	171	171	26.488	26.723	0.235	[-0.236, 0.791]
NGBoost	15	10-49	2433	2433	19.786	19.595	-0.191	[-0.412, 0.048]
NGBoost	15	50+	2294	2294	68.290	60.217	-8.072	[-10.457, -5.269]
NGBoost	60	0	26	26	43.258	43.478	0.220	withheld
NGBoost	60	1-2	20	20	20.559	19.444	-1.114	withheld
NGBoost	60	3-9	51	51	23.298	22.043	-1.255	[-2.645, 0.086]
NGBoost	60	10-49	1585	1585	14.874	15.067	0.193	[-0.034, 0.417]
NGBoost	60	50+	3318	3318	40.950	37.395	-3.554	[-4.772, -2.300]

S10.3 Empirical references by prefix stratum

Source	min	Bin	Roots	Units	Unconditional	Conditional	Difference
Weibo	5	0	1622	1438	69.613	68.799	-0.814
Weibo	5	1-2	1516	1277	48.962	49.135	0.173
Weibo	5	3-9	1251	934	97.663	96.250	-1.414
Weibo	5	10-49	526	362	363.970	285.892	-78.078
Weibo	5	50+	85	67	2408.482	1860.098	-548.384
Weibo	15	0	764	708	69.123	69.922	0.799
Weibo	15	1-2	982	889	58.278	54.237	-4.041
Weibo	15	3-9	1720	1357	29.640	26.937	-2.703
Weibo	15	10-49	1168	816	107.482	96.351	-11.131
Weibo	15	50+	366	260	1025.623	733.616	-292.007
Weibo	60	0	334	316	77.333	77.576	0.243
Weibo	60	1-2	386	362	24.332	23.179	-1.153
Weibo	60	3-9	1547	1312	28.344	24.186	-4.158
Weibo	60	10-49	1814	1341	28.115	25.949	-2.166
Weibo	60	50+	919	617	436.394	314.331	-122.063
Seismic	5	0	103	103	38.652	36.236	-2.416
Seismic	5	1-2	106	106	52.540	50.350	-2.189
Seismic	5	3-9	497	497	26.365	20.556	-5.809
Seismic	5	10-49	2866	2866	32.362	27.728	-4.634
Seismic	5	50+	1428	1428	156.917	114.546	-42.372
Seismic	15	0	53	53	40.056	34.660	-5.396
Seismic	15	1-2	49	49	41.346	39.669	-1.677
Seismic	15	3-9	171	171	29.260	25.681	-3.579
Seismic	15	10-49	2433	2433	25.449	19.396	-6.053
Seismic	15	50+	2294	2294	94.561	70.302	-24.259
Seismic	60	0	26	26	55.260	42.323	-12.938
Seismic	60	1-2	20	20	31.135	20.474	-10.661
Seismic	60	3-9	51	51	29.828	22.795	-7.033
Seismic	60	10-49	1585	1585	20.031	14.149	-5.882
Seismic	60	50+	3318	3318	54.417	41.304	-13.113

S11 Complete count-calibration graphics

Both PIT constructions and all window/family panels are shown. The saved `CALIBRATION.json` file supplies bin values for every procedure, including the simple references. No recalibration was fitted.

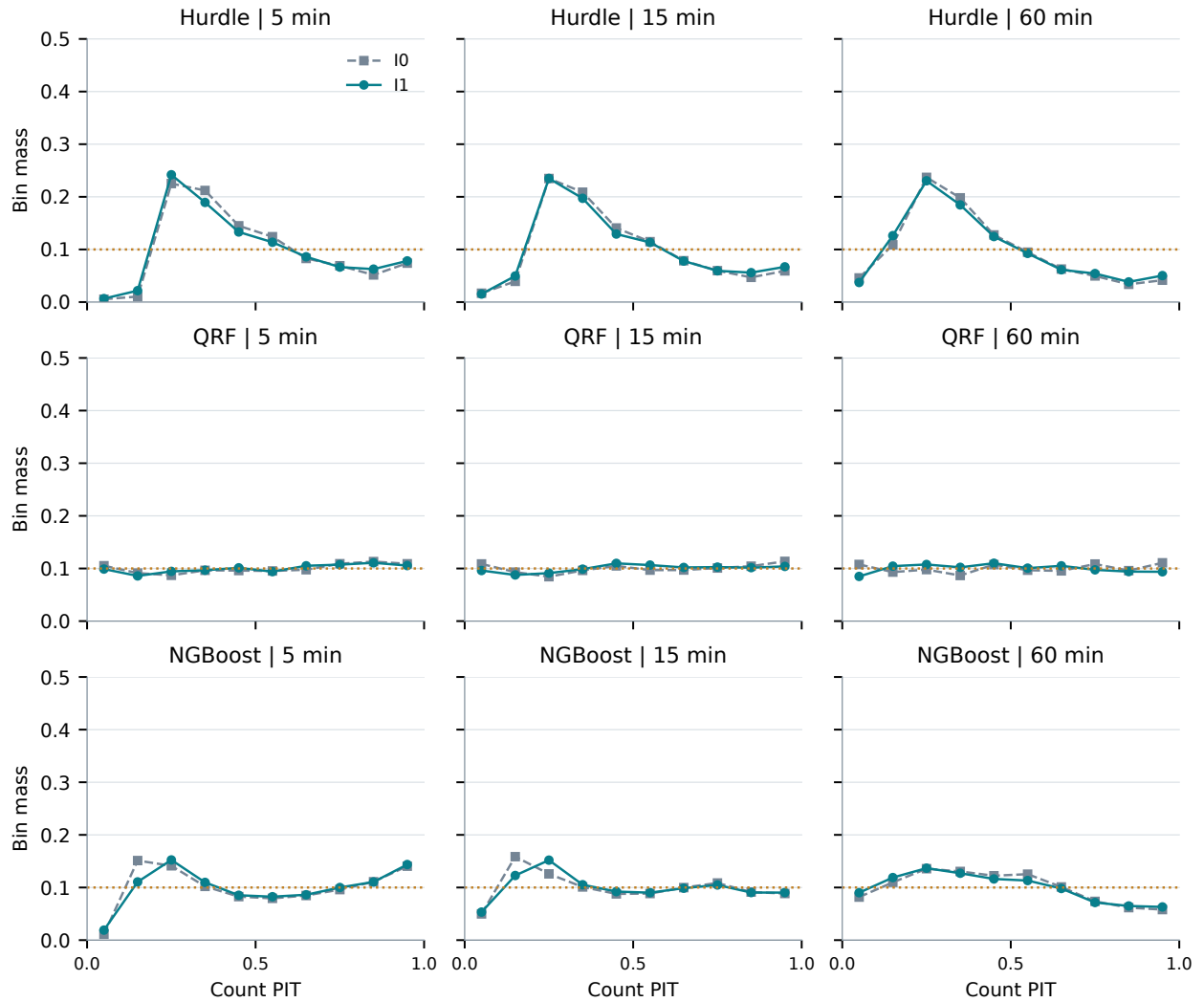


Figure S4: Weibo: nonrandomized count PIT by family and prefix. The horizontal dotted line is 0.1. Seeds are averaged as prescribed; the panels are descriptive and do not add independent test cases.

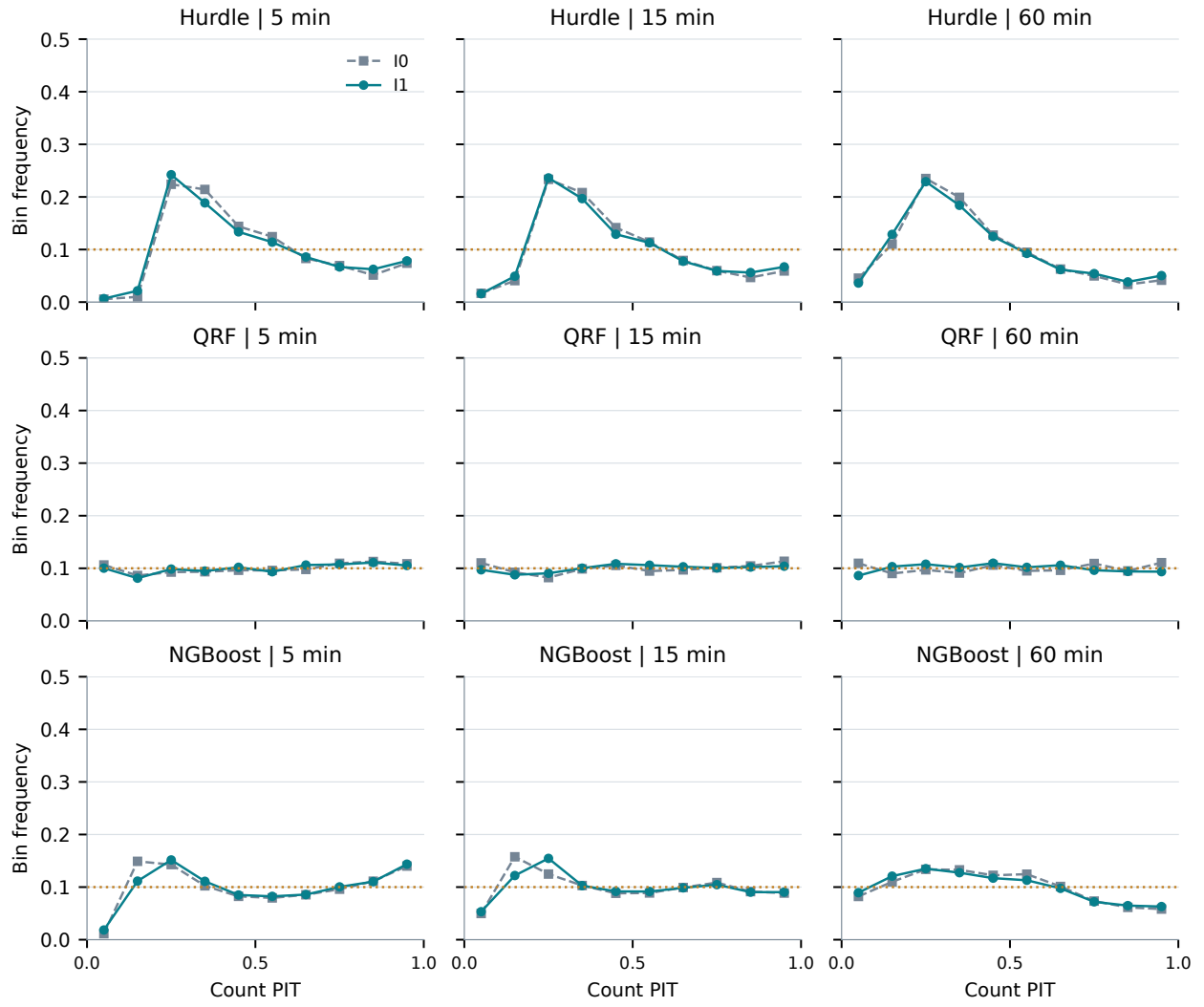


Figure S5: Weibo: randomized count PIT by family and prefix. The horizontal dotted line is 0.1. Seeds are averaged as prescribed; the panels are descriptive and do not add independent test cases.

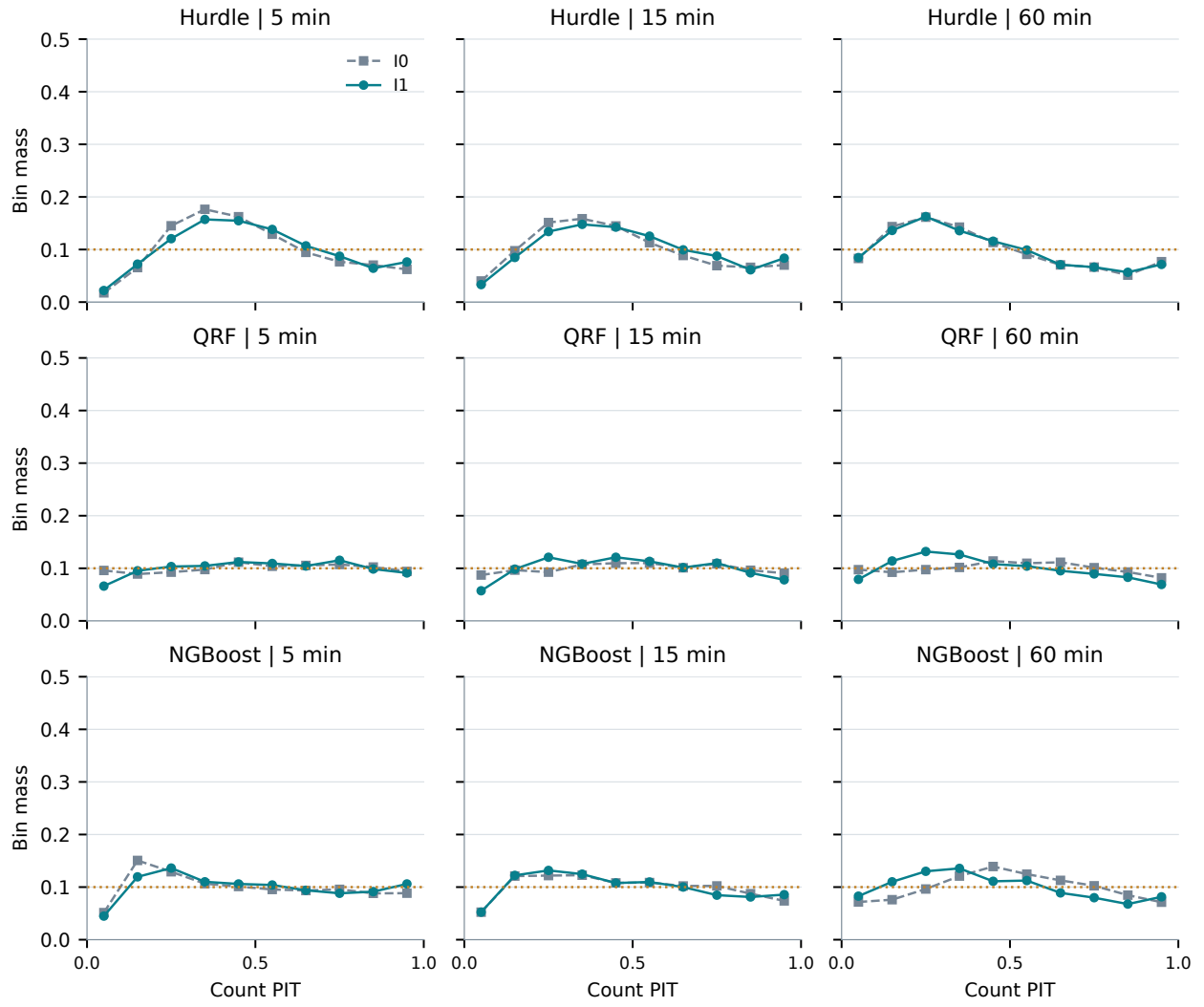


Figure S6: SEISMIC: nonrandomized count PIT by family and prefix. The horizontal dotted line is 0.1. Seeds are averaged as prescribed; the panels are descriptive and do not add independent test cases.

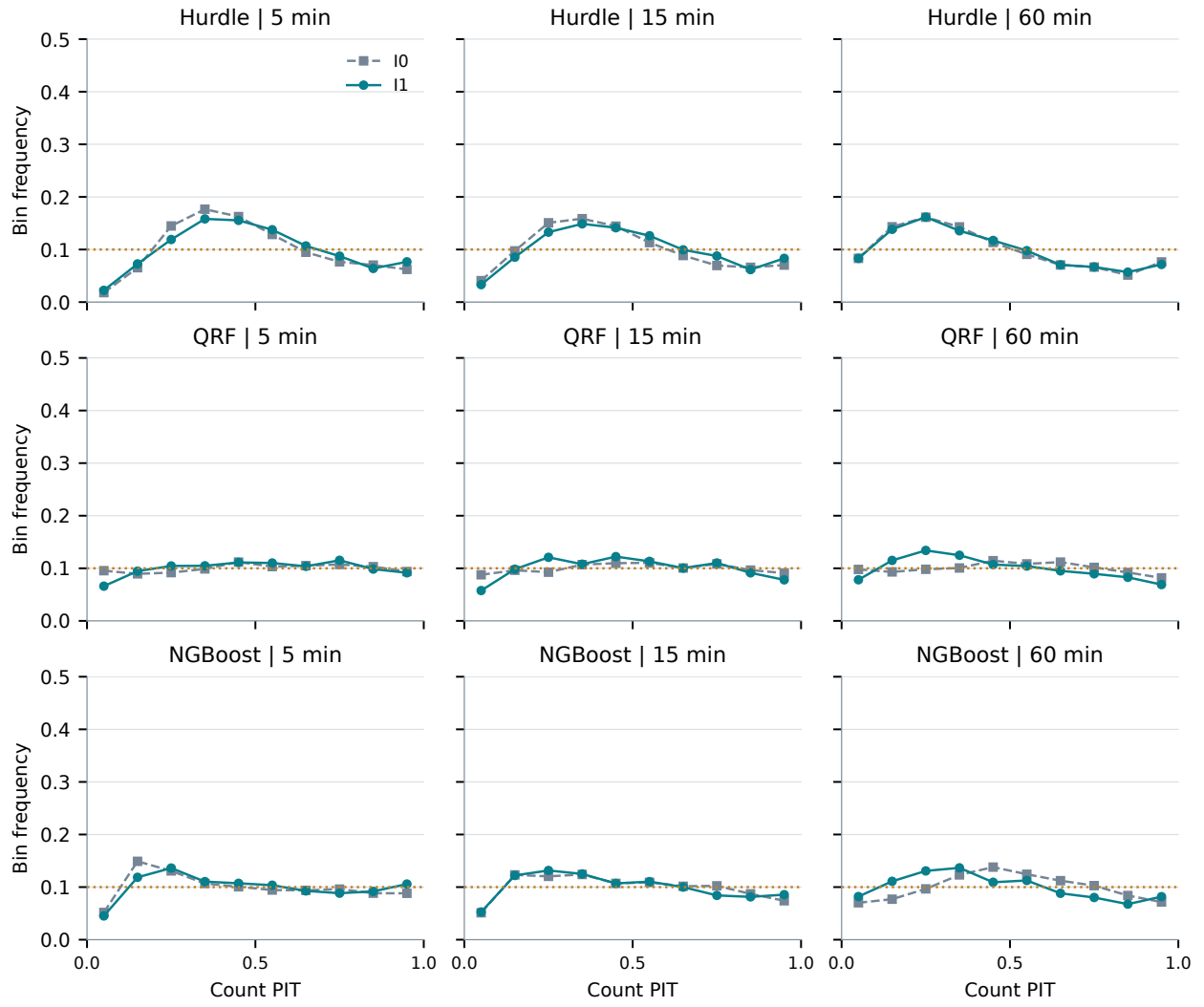
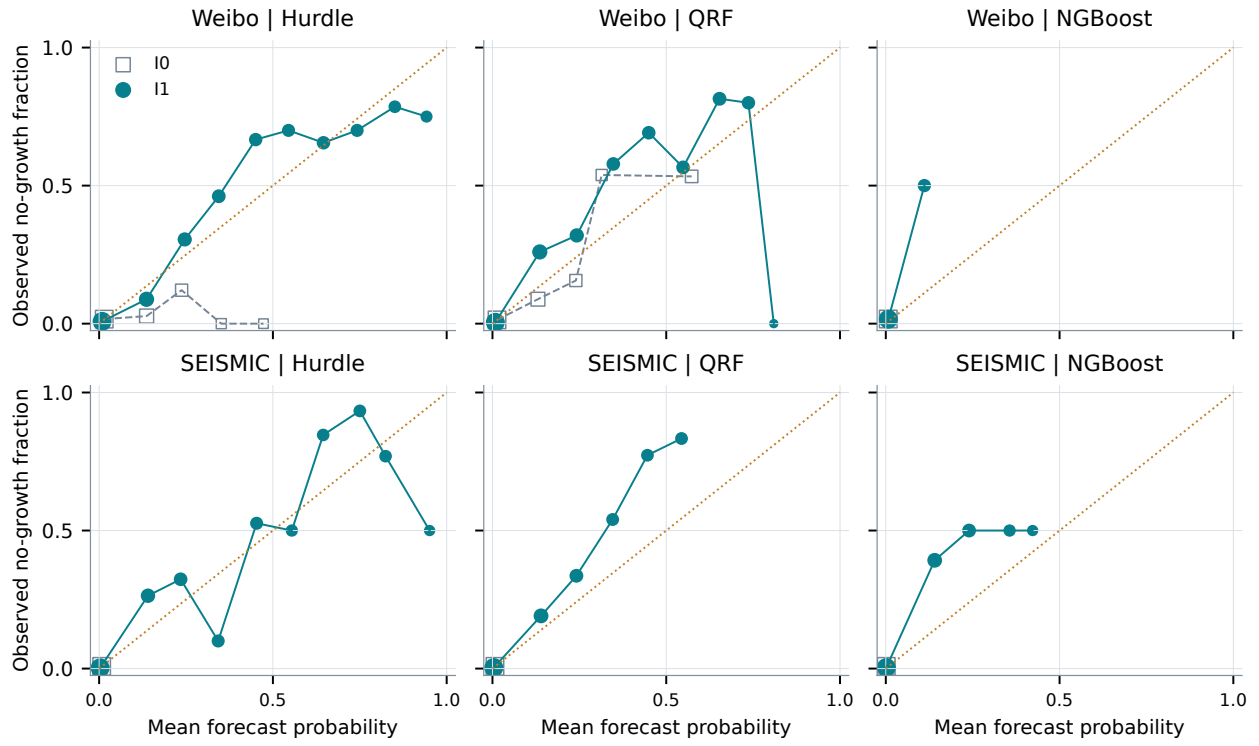


Figure S7: SEISMIC: randomized count PIT by family and prefix. The horizontal dotted line is 0.1. Seeds are averaged as prescribed; the panels are descriptive and do not add independent test cases.



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